When Does Multi-Modal Fusion Stay Robust? A ROS2 Study of Fusion Strategies for Tactile Material Classification under Sensor Noise

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Sensor: Daimon S2508080077 (GelSight-style optical tactile sensor)

***Abstract*—A single optical tactile sensor of the GelSight family exposes, at every contact, four synchronised views of the same event: a raw gel image, a depth map, an in-plane deformation field, and a tangential shear field. Fusing these views is known to improve material recognition, but *which* fusion strategy should be used, and *whether that choice is stable when the sensor becomes unreliable*, is far less clear. We study three canonical fusion strategies—input-level (early), decision-level (late), and feature-level (hybrid)—against four single-modality baselines, using an identical shared encoder for a fair comparison, on a six-class fruit classification task (245 trials) recorded with a Daimon sensor. The evaluation is driven through a ROS2 (`rclpy`) pipeline that replays the recorded test trials and injects *synthetic* perturbations (relative Gaussian noise and modality dropout) under explicit assumptions, sweeping six noise levels over five seeds. On clean data early and hybrid fusion tie for the best accuracy (0.837), and both beat every baseline. Under noise, however, the two diverge sharply: early fusion is essentially noise-invariant (0.837→0.857), whereas hybrid fusion is the single most fragile model, collapsing below chance (0.837→0.110); late fusion degrades steadily in between. A paired t-test confirms early fusion's advantage over the strongest baseline is highly significant ($t = 74.9$, $p \approx 6 \times 10^{-31}$). We give a mechanistic explanation grounded in *where* corruption enters the network, and map the pipeline onto the COGAR cognitive design-pattern language.**

***Index Terms*—tactile sensing, GelSight, multi-modal fusion, material classification, sensor noise, robustness, ROS2, cognitive architectures**

I. Introduction and Research Question

For a robot to manipulate the physical world it must perceive the materials it touches. Optical tactile sensors of the GelSight family convert contact into a high-resolution, image-like signal, and a single such sensor can simultaneously expose several complementary descriptions of one contact event. This makes them a natural test-bed for *multi-modal fusion*: the four streams describe the same physical interaction from different angles, and combining them is widely reported to help recognition [1], [3].

The usual question—*does fusion help?*—is by now largely answered in the affirmative. We ask a sharper and more practical one. A deployed sensor is never perfectly clean: gel wear, illumination drift, calibration error, and electronic noise all degrade one or more streams, and they rarely degrade them equally. The decision a system designer actually faces is therefore not *whether* to fuse but *how*, and crucially *whether the best fusion strategy remains best as the sensor becomes unreliable*.

Research question. Following the Intervention–Population–Outcome–Comparison framing of experimental design, we phrase the study as:

> *For tactile material classification from a single four-modality optical sensor (**population**), how does the choice of fusion strategy—early, late, or hybrid (**intervention**)—affect classification accuracy and its robustness to sensor noise (**outcome**), compared with single-modality baselines and with one another (**comparison**)?*

The study yields one main finding: **on clean data the learned fusions are indistinguishable, but their robustness to noise is not—and the ranking among them is the opposite of what a "more structure is better" intuition would suggest.** Input-level (early) fusion, which entangles the modalities before committing to any learned feature, is almost noise-invariant; feature-level (hybrid) fusion, which processes each modality in isolation before combining, is the most fragile of all seven models. Decision-level (late) fusion lies in between. The optimal fusion strategy is therefore *contingent on sensor reliability*, not a fixed property of the data.

II. Literature Analysis

Optical tactile sensing and material perception. GelSight-style sensors image the deformation of a soft gel pressed against an object, yielding a tactile signal amenable to computer-vision methods. Yuan *et al.* [1] collected colour, depth, and tactile data for many fabrics and jointly trained CNNs across modalities, showing that a model trained on vision *and* touch can outperform a vision-only model—direct evidence that the streams carry complementary, material-discriminative information. Our raw-image and depth modalities are the single-sensor analogue of that visual/geometric channel.

Visual–tactile fusion for manipulation. Calandra *et al.* [3] fuse vision and touch in an end-to-end multi-modal CNN to predict grasp outcomes, motivating fusion at the level of learned representations rather than independent per-modality decisions—the same distinction that separates our early/hybrid models from late fusion.

TABLE I
THE FOUR SYNCHRONISED TACTILE MODALITIES.

Modality	Channels	Semantic content
rawimg	1	grayscale gel image (surface micro-geometry)
depth	1	per-pixel z-displacement (compliance)
deformation	2	in-plane (x, y) displacement field
shear	2	tangential (x, y) displacement field

Adaptive fusion. Ding *et al.* [2] propose an adaptive visual–tactile network that *learns to weight* modalities, including dropout-based reliability handling, for recognition of multi-material systems. This is precisely the mechanism our results implicate: when modality reliability is imbalanced, the way a method handles an unreliable stream determines its robustness. Our work can be read as an empirical isolation of *why* the locus of fusion matters once reliability is no longer guaranteed.

Surveys. The breadth of tactile multi-modal fusion is catalogued in recent surveys [4], which organise approaches by the level at which modalities are combined. Our early/late/hybrid taxonomy is a concrete instance of that input/decision/feature-level distinction, evaluated on an all-tactile, four-modality stream from one sensor and—unlike most prior work—stress-tested under controlled sensor degradation.

III. METHODOLOGICAL PROPOSAL

At a conceptual level the system is a perception pipeline that turns one contact event into a material label. It has three conceptual stages: a sensing stage that produces four modality streams, a fusion stage that combines them, and a decision stage that emits a class. The experimental contribution is to vary *only* the fusion stage while holding everything else fixed, and then to observe how each variant behaves as the sensing stage is progressively corrupted.

The four modalities (Table I) describe the same contact complementarily. Depth and deformation encode *compliance*: rigid fruits push deep, soft fruits displace little. The raw image encodes *surface micro-geometry*: the dimpled imprint of an orange, the smooth imprint of an apple. Shear encodes *tangential interaction*; under a deliberately static pressing protocol it carries little class information and is the weakest stream.

Three fusion strategies, each a different composition of the same building blocks:

- *Early (input-level)* fusion stacks all modalities into a single multi-channel input and processes them with *one* shared encoder. Modalities are entangled from the first layer.
- *Late (decision-level)* fusion trains one independent classifier per modality and averages their decisions with equal, fixed weight. It has no trainable cross-modal parameters.
- *Hybrid (feature-level)* fusion processes each modality with its *own* encoder, then concatenates the resulting feature vectors and learns a small head over them. Modalities are combined only after each has been independently committed to a high-level feature.

A single compact convolutional encoder is reused across all variants so that differences in performance reflect the *fusion topology*, not encoder capacity.

Architectural mapping (COGAR). The pipeline maps cleanly onto the course design-pattern language. The Daimon sensor and its driver form a *Sensor-Device* that hides the hardware and exposes four streams; each stream is conceptually a *topic* in a publish–subscribe system; normalisation and peak-frame selection are stateless *Adapters*; each encoder and fusion head is a *Computational* component; and the whole forms a *Pipeline* whose intermediate per-modality outputs are reusable—exactly what makes hybrid fusion expressible. The empirical contribution can then be restated architecturally: *the composition topology, not just the internal model, determines robustness.*

IV. EXPERIMENT DESIGN

A. Dataset

Six fruit classes were recorded with a Daimon S2508080077 sensor: apple, orange, kiwi, banana, nectarine, and peach (245 valid trials; nectarine is a fuzzless near-duplicate of peach, making the task realistically hard). Each trial records 30 frames ($\sim$1 s) of a controlled contact. Pressure was deliberately varied within each class to encourage force-robust features rather than memorisation of one contact. For every trial the single *peak-deformation* frame—the moment of strongest contact—is selected as the most informative sample. One kiwi trial with near-zero signal (a missed contact) was excluded.

B. Models and training

Seven models are trained identically: four single-modality baselines and the three fusion strategies. All reuse the same $\sim$0.4M-parameter convolutional encoder. Training uses AdamW (lr 10^{-3}, weight decay 10^{-4}), a cosine schedule, and cross-entropy loss; the best checkpoint by validation accuracy is retained and the test set is used once. Splits are constructed at the *trial* level with a fixed seed (42) to prevent frame leakage, giving the identical test trials used throughout.

C. ROS2 evaluation pipeline

Evaluation is performed through a ROS2 (`rclpy`) pipeline rather than an offline script, so that the experiment is reproducible and controllable as a running system:

- A **replay publisher** (Sensor-Device) loads each recorded test trial, selects its peak frame, optionally perturbs it, and publishes the four modalities on separate topics.
- A **classifier subscriber** (Computational) re-assembles each trial from the four topics, runs all seven models, and publishes the per-model predictions.
- A **Jupyter notebook** drives the pipeline through `ipywidgets`: it sets the perturbation parameters, runs the full test split for each condition, logs predictions to disk, and aggregates the results.

TABLE II
CLEAN SIX-CLASS TEST RESULTS ($\ell = 0$). RANDOM BASELINE $= 0.167$.

Model	Accuracy	Macro-F_1
early_fusion	**0.837**	0.836
hybrid_fusion	**0.837**	0.837
late_fusion	0.735	0.721
rawimg	0.694	0.687
depth	0.694	0.697
deformation	0.633	0.628
shear	0.469	0.471

D. Synthetic perturbations and assumptions

The *base* data are real recorded contacts; only the perturbations are synthetic, and they are stated explicitly:

- **A1** *Relative Gaussian noise*: each modality is corrupted by zero-mean Gaussian noise whose standard deviation scales with that modality's own spread, $\sigma = \ell \cdot (\mathrm{std}(\text{modality}) + \varepsilon)$, so that the dimensionless *noise level* ℓ corrupts every stream by a comparable fraction. $\ell = 0$ reproduces the clean results.
- **A2** *Modality dropout*: a whole modality is set to zero, emulating a dead sensor stream while the others continue.
- **A3** *Single specimen per class*: every model is evaluated on the *same* perturbed data, so the *relative* fusion comparison—our actual claim—is insulated from absolute-accuracy effects.

E. Protocol and metrics

The noise level is swept over $\ell \in \{0.0, 0.1, 0.2, 0.3, 0.4, 0.5\}$. Because a single run is not representative, each $\ell > 0$ is repeated over five seeds and reported as mean $\pm$ std; $\ell = 0$ is deterministic and run once. The metrics are test accuracy and macro-F_1 (aligned with the multi-class recognition question), and the noise-robustness curve (accuracy versus ℓ). Statistical significance of the best fusion model over the best baseline is assessed with a *paired* t-test, pairing on the identical (ℓ, seed) scenarios. The full configuration (seeds, noise levels, split seed, software stack) is fixed and logged for reproducibility.

V. RESULTS AND VISUALIZATION

A. Clean data: fusion beats baselines

On clean data (Table II, Fig. 1) both learned fusions reach 0.837 accuracy, ahead of every single-modality baseline; the strongest baselines (rawimg, depth) reach only 0.694. Late fusion (0.735) also beats the baselines but trails the learned fusions. The raw image is among the strongest single modalities despite identical mean brightness across classes, confirming that the discriminative information is *spatial*; shear is weakest, as expected under static contact. The confusion matrices (Fig. 2) show the dominant residual errors are banana$\to$orange and apple$\to$nectarine—contact-patch similarities under our protocol—not the botanical peach/nectarine pair, which the strong models separate well.

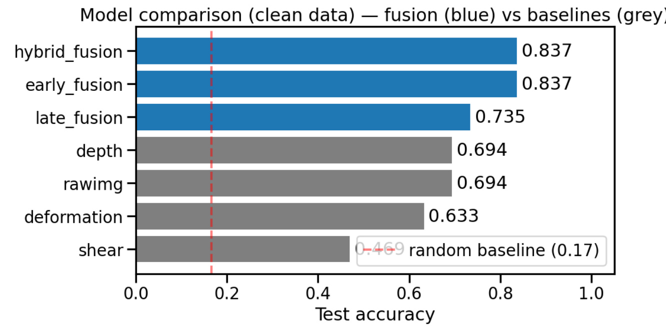

Fig. 1. Clean-data model comparison: fusion (blue) versus single-modality baselines (grey). Early and hybrid fusion are indistinguishable here.

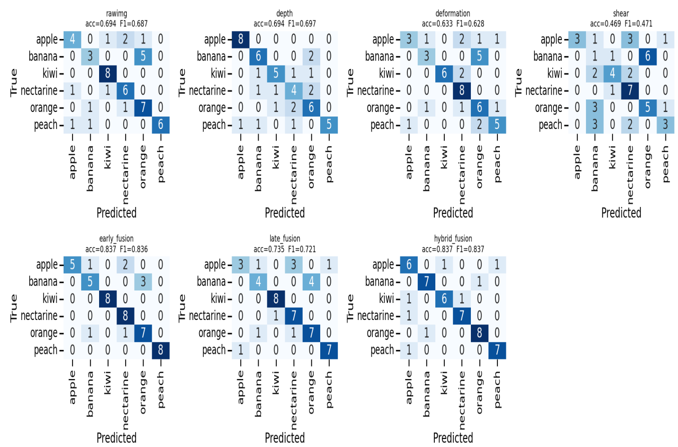

Fig. 2. Clean-data confusion matrices for all seven models. The learned fusions (bottom-left, bottom-right) have the cleanest diagonals.

B. Noise robustness: the strategies diverge

The clean-data tie is misleading. Table III and Fig. 3 report accuracy as the relative noise level rises. Three behaviours stand out.

Early fusion is noise-invariant. Its accuracy is essentially flat and even edges up slightly ($0.837 \to 0.857$ at $\ell = 0.5$), with negligible seed-to-seed variance.

Hybrid fusion is the most fragile model of all seven. It matches early fusion at $\ell = 0.1$ (0.845) but then collapses—0.714, 0.380, 0.192, 0.110—ending *below* the 0.167 random baseline. Its degradation (-0.727) is the largest in the study.

Late fusion degrades steadily ($0.735 \to 0.335$, -0.400), gracefully relative to hybrid but far from robust.

Among the baselines, depth is the most fragile (it collapses to chance already at $\ell = 0.1$), rawimg degrades steadily, and—perhaps surprisingly—deformation is the most robust single stream, staying near 0.63 throughout: its vector-field structure is comparatively resilient to per-pixel Gaussian noise.

C. Statistical significance

Averaged across the whole sweep, early fusion (mean 0.853) is the best model and deformation (mean 0.624) the best baseline. A paired t-test over the 26 identical (ℓ, seed) scenarios rejects the null hypothesis of equal means decisively: mean

TABLE III
NOISE-ROBUSTNESS: MEAN ACCURACY ACROSS FIVE SEEDS AT EACH RELATIVE NOISE LEVEL ℓ. BEST PER COLUMN IN **BOLD**.

Model	0.0	0.1	0.2	0.3	0.4	0.5
early_fusion	**.837**	.841	**.857**	**.857**	**.857**	**.857**
hybrid_fusion	**.837**	**.845**	.714	.380	.192	.110
late_fusion	.735	.788	.743	.559	.453	.335
rawimg	.694	.531	.327	.196	.163	.163
depth	.694	.163	.163	.163	.163	.163
deformation	.633	.633	.633	.612	.612	.629
shear	.469	.461	.449	.449	.449	.469

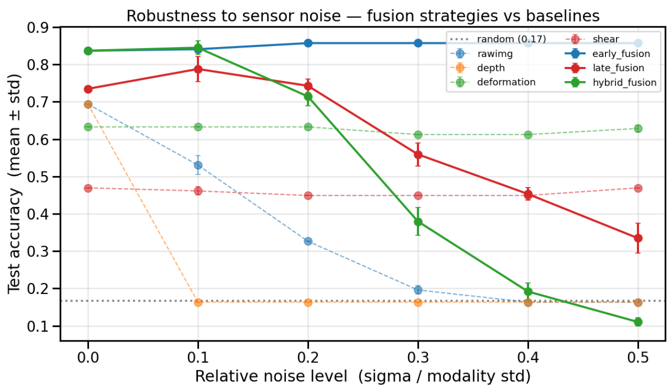

Fig. 3. Accuracy versus relative noise level (mean ± std over five seeds). Early fusion (solid blue) is flat; hybrid fusion (solid green) starts tied with it then collapses below chance; late fusion (solid red) declines steadily.

difference $+0.229$, $t = 74.9$, $p \approx 6 \times 10^{-31}$. Early fusion is also significantly more robust than the other two fusion strategies across the sweep (vs. hybrid: $t = 6.6$, $p \approx 7 \times 10^{-7}$; vs. late: $t = 7.7$, $p \approx 5 \times 10^{-8}$).

VI. DISCUSSION AND LIMITATIONS

A. *Why early fusion stays robust and hybrid fusion fails*

The clean-data tie and the noise-time divergence have a single, clean explanation rooted in *where* corruption enters the network.

In **early fusion** the modalities are stacked at the input and processed by one shared encoder. Noise enters before any learned feature is formed, and the very first convolution mixes all channels within each receptive field. When one channel (e.g. depth) is corrupted, the same filters still receive strong signal from the others, so the network can route around the bad channel; cross-channel redundancy and spatial smoothing suppress the noise early. The result is a model that barely notices the corruption.

In **hybrid fusion** each modality is processed in *isolation* by its own deep encoder before any cross-modal information is available. A corrupted depth input is therefore transformed, through the full depth encoder, into a confidently *wrong* high-level feature, which is concatenated into the fused vector. The downstream head—trained on clean data to trust the depth feature—weights this garbage heavily and has no way, for a given sample, to discover that depth is unreliable. The collapse of the standalone depth baseline (-0.531) thus *propagates* into hybrid fusion (-0.727). Isolation, which looks like a virtue on clean data, becomes the failure mode under noise.

Late fusion sits between the two: a corrupted modality contributes only one of four equally weighted votes, so the damage is bounded to roughly a quarter of the decision. This caps the degradation (more graceful than hybrid) but, with fixed equal weights, cannot suppress the bad stream either (far from early fusion's robustness).

The general principle is that *the locus of fusion governs robustness*: the earlier modalities are entangled, the more a network can exploit redundancy to absorb the failure of any single stream; the later they are combined, the more a single corrupted stream can commit the system to a wrong answer. This refines the common intuition that more elaborate fusion is uniformly better—under realistic sensor degradation the simplest learned fusion was the most robust.

B. *Physical interpretation*

A capable classifier must combine compliance cues (depth, deformation) with surface cues (the raw image); fusion exploits exactly this complementarity. The robustness ordering is consistent with the physics: depth confounds pressure with material and, being a single scalar field, offers no redundancy to fall back on when corrupted—hence its immediate collapse and its toxic effect inside hybrid fusion. Deformation, a smooth two-channel vector field, retains usable structure under per-pixel noise. Shear is intrinsically weak under static contact and so is neither helped nor hurt much by noise.

C. *Relation to task difficulty*

The present noise study complements an earlier observation from the project's data on the same sensor, where decision-level fusion was strongest on an easy three-class task but weakest on the hard six-class task while learned fusion stayed strong. Taken together, the two analyses point to the same conclusion from two directions: *there is no universally optimal fusion strategy*. The best choice depends jointly on task difficulty and on the expected reliability distribution of the sensors—and a system designer who must tolerate sensor degradation should prefer input-level fusion.

D. *Limitations*

Single specimen per class: all trials of a class come from one physical fruit, so absolute accuracies may overstate true material-level generalisation; because all models share the same split and the same perturbations, the *relative* comparison is insulated, but a specimen-stratified test is the proper next step. *Modest test set* (49 samples): one prediction shifts accuracy by $\sim$2%, so small clean-data gaps should be read cautiously—though the noise-time divergence is far too large to be sampling noise. *Single training seed*: robustness to training stochasticity was not swept here. *Noise model* (A1): additive, independent, zero-mean Gaussian noise is a deliberate simplification; structured degradations (motion blur, gel wear, miscalibration) may rank the strategies differently. *Static* *protocol*: a brief controlled slide would activate shear and could change the fusion ordering.

E. Future work

The same ROS2 classifier node is designed to be re-pointed from the replay publisher to a *live* Daimon sensor publisher, turning the offline study into an online deployment in which the sensor streams the four modalities in real time and the node emits predictions continuously. Combined with a multi-seed, specimen-stratified protocol and richer (non-Gaussian) degradations, this would test whether the robustness advantage of input-level fusion observed here holds on a physically deployed system.

VII. Conclusion

We compared three fusion strategies and four baselines for tactile material classification on a Daimon optical sensor, evaluating them through a ROS2 pipeline that injects synthetic sensor noise under explicit assumptions. On clean data the two learned fusions are indistinguishable and both beat every baseline. Under noise they diverge completely: input-level (early) fusion is essentially noise-invariant, decision-level (late) fusion degrades steadily, and feature-level (hybrid) fusion—tied for best when clean—is the most fragile model of all, collapsing below chance. The advantage of early fusion over the strongest baseline is statistically overwhelming. The locus of fusion, not its sophistication, governs robustness: entangling modalities early lets a network absorb the failure of any single stream, whereas committing each modality to an isolated feature lets one corrupted stream dominate. The best fusion strategy is thus contingent on sensor reliability, and a noise-tolerant tactile system should fuse at the input.

References

[1] W. Yuan, S. Wang, S. Dong, and E. H. Adelson, "Connecting look and feel: Associating the visual and tactile properties of physical materials," in *Proc. IEEE CVPR*, 2017, pp. 4494–4502.

[2] Z. Ding, G. Chen, Z. Wang, and L. Sun, "Adaptive visual–tactile fusion recognition for robotic operation of multi-material systems," *Front. Neurorobot.*, vol. 17, art. 1181383, 2023.

[3] R. Calandra, A. Owens, D. Jayaraman, J. Lin, W. Yuan, J. Malik, E. H. Adelson, and S. Levine, "More than a feeling: Learning to grasp and regrasp using vision and touch," *IEEE Robot. Autom. Lett.*, vol. 3, no. 4, pp. 3300–3307, 2018.

[4] "Tactile-based multimodal fusion in embodied intelligence: A survey of vision, language, and contact-driven paradigms," arXiv:2605.17336, 2026.